

COURSE TITLE : SURVEY PRACTICAL FOR ARCHITECTURE
COURSE CODE : 3188
COURSE CATEGORY : B
PERIODS/WEEK : 3
PERIODS/SEMESTER : 45
CREDITS : 1

TIME SCHEDULE

Module	Topic	Periods
1	Surveying instruments	9
2	Levelling	15
3	Theodolite Surveying	12
4	Total Station	9
TOTAL		45

Sl.	Sub	Student will be able to
1	1	Perform survey in the field using chain, tape , ranging rod, cross staff and peg for calculate an area of small plot.
	2	Identify the plane table and accessories.
	3	Perform survey in the field by chain and compass
2	1	Conduct levelling practice
	2	Perform contouring survey
	3	Perform theodolite survey in the field.
3	1	Find the difference in elevations and distance between objects
	2	Perform Total station survey in the field.

Module I

1.1.0 Perform survey in the field using chain, tape , ranging rod, cross staff and peg for calculate an area of small plot.

- 1.1.1 Select convenient observation stations
- 1.1.2 Perform survey on the field and Record the observations in the field book
- 1.1. To calculate the area from the given field data.

1.2.0 Identify the plane table and accessories.

1.3.0 Perform survey in the field by chain and compass

- 1.3.1 Perform the survey work using compass and accessories, record in the field book
- 1.3.2 To calculate the inaccessible distance by compass and chain.

Module II

2.1.0 Conduct levelling practice

- 2.1.1 Perform temporary adjustments for taking observations
- 2.1.2 Take fly levels for establishing reduced level for a bench mark
- 2.1.3 Determine the reduced level of given points.

2.2.0 Perform contouring

- 2.2.1 Take spot levels for preparing contour map.
- 2.2.2 Prepare contour map.

Module III

3.1.0 Perform theodolite survey in the field.

- 3.1.1 Perform temporary adjustments
- 3.1.2 Measurements of horizontal angles- different methods-vertical angles.
- 3.1.3 Record the observation in the field book

3.2.0 Find the difference in elevations and distance between objects.

- 3.2.1 Establish a base line
- 3.2.2 Determine heights and distances of the objects by observations from a base line.
- 3.2.3 Determine difference in level and horizontal distance between two points by observation from a single station; also from the ends of the base line.

Module IV

4.1.0 Perform Total station survey in the field.

4.1.1 Component parts and features of a Total Station

4.1.2 Setting up the Total Station

4.1.3 Measurement of Distance, Angle, Bearing etc.

4.1.4 Field procedure for co-ordinate measurement

4.1.5 Field procedure to run a traverse survey

4.1.6 Downloading data from Total Station Instrument and Processing with software

COURSE CONTENTS

MODULE I

Survey in the field using chain - tape - ranging rod - cross staff and peg - calculate an area of small plot - select convenient observation stations - Perform survey on the field and Record the observations in the field book - calculate the area from the given field data - plane table and accessories - field survey by chain and compass - survey work using compass and accessories - record in the field book - calculate the inaccessible distance by compass and chain.

MODULE II

Contouring - Take spot levels for preparing contour map - Prepare contour map.

MODULE III

Find the difference in elevations and distance between objects -Establish a base line - Determine heights and distances of the objects by observations from a base line - Determine difference in level and horizontal distance between two points by observation from a single station - also from the ends of the base line.

MODULE IV

Total station survey in the field - Component parts and features of a Total Station - Setting up the Total Station - Measurement of Distance, Angle, Bearing etc.- Field procedure for co-ordinate measurement - Field procedure to run a traverse survey -Downloading data from Total Station Instrument and Processing with software .

REFERENCE BOOKS

1. T.P.Kanetkar & Kulkarni : Surveying and Levelling (Voll&Vol II) ; Jain book depot
2. B.C.Punmia : Surveying – II ; Laxmipublications
3. K.R Arora : Surveying - ; Standard Book House
4. P.B.Shahai : A textbook of surveying [Vol.I and Vol. II] ; Oxford and IBH Publishing Co.
5. Patel .A.N : Remote Sensing Principles & Applications ; Scientific Publishers
6. Lillesand : Remote sensing and image interpretation ; Wiley

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|---------------------|---|--------------------------------------|
| 7. SathishGopi | : Advanced Surveying | ; Pearson |
| 8. CL Kocher | : Text book of Surveying | ; Dhanpat Rai Publishing Co (p) |
| 9. N N Basak | : Surveying | ; TataMcgraw Hill |
| 10. S.S.Bhavikatti | : Surveying and leveling (Vol. I &II) | ; I.K.International publishing house |
| 11. NITTTR, Chennai | : AICTE Continuing Education module on Geographical information systems | |